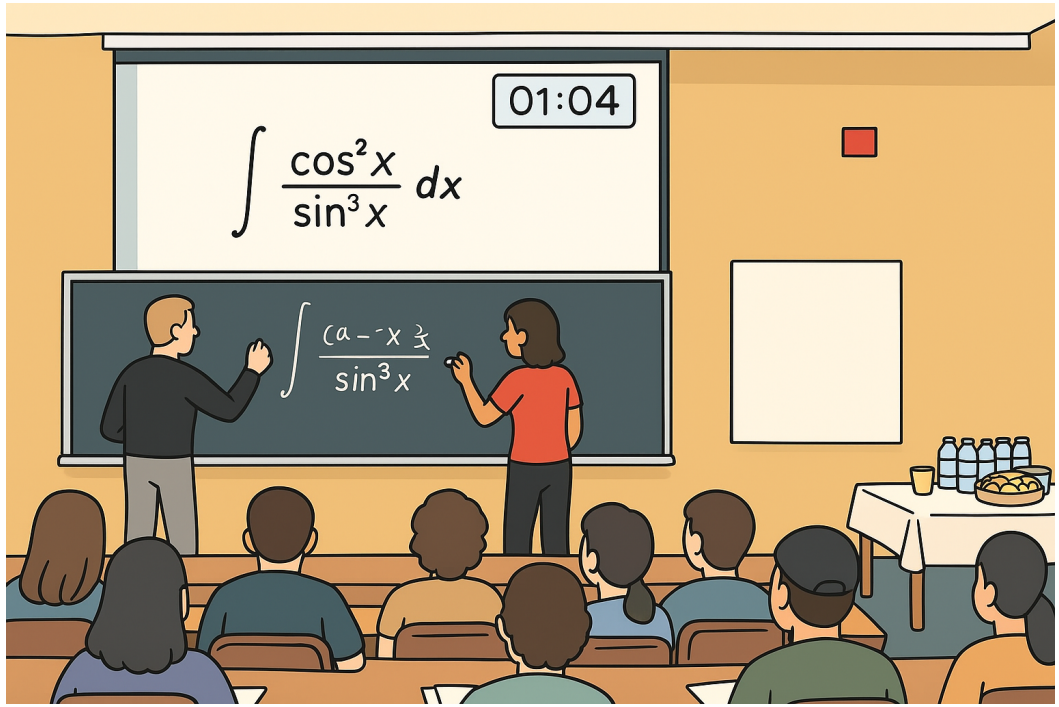
Integratión Bee Physics
Muerte Subita



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

Muerte Súbita

Integral #34 **Integratión Bee Physic**

$$\int \frac{\tan^{-1}\sqrt{x}}{\sqrt{x}(1+x)}\,\mathrm{d}x$$

$$\int \frac{\tan^{-1}\sqrt{x}}{\sqrt{x}(1+x)}\,\mathrm{d}x =$$

$$\left(\arctan\sqrt{x}\right)^2 + C$$

Integral #5 Integratión Bee Physic

$$\int \frac{x}{1+e^{x^2}}\,\mathrm{d}x$$

Respuesta Integral #5

Integratión Bee Physic

$$\int \frac{x}{1+e^{x^2}}\,\mathrm{d}x =$$

$$\frac{1}{2}\ln\left(1+e^{x^2}\right)+\mathrm{C}$$

Integral #80 **Integratión Bee Physic**

$$\int_0^1 x^{2024}\left(1-x^{2025}\right)^{2025}\,\mathrm{d}x$$

Respuesta Integral #80 **Integratión Bee Physic**

$$\int_0^1 x^{2024}\left(1-x^{2025}\right)^{2025}\,\mathrm{d}x = \frac{1}{(2025)(2026)}$$

Integral #91 **Integratión Bee Physic**

$$\int \frac{1}{\cosh^2 x}\,\mathrm{d}x$$

$$\int \frac{1}{\cosh^2 x}\,\mathrm{d}x =$$

$$\tanh x + C$$

Integral #88 **Integratión Bee Physic**

$$\int \cos(\arctan x)\,\mathrm{d}x$$

$$\int \cos(\arctan x)\,\mathrm{d}x =$$

$$\ln\left|\sqrt{1+x^2}+x\right| + C$$

Integral #23 **Integratión Bee Physic**

$$\int \frac{\sin^2 x}{\cos^6 x}\,\mathrm{d}x$$

$$\int \frac{\sin^2 x}{\cos^6 x} \mathrm{d}x =$$

$$\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} + C$$

Integral #24 **Integratión Bee Physic**

$$\int \left\{ \sqrt[9]{\frac{\sqrt[5]{x^2} \cdot \sqrt[5]{x^2} \cdots (100 \text{ veces})}{\sqrt[4]{x} \cdot \sqrt[4]{x} \cdots (340 \text{ veces})}} \right\} \mathrm{d}x$$

$$\int \left\{ \sqrt[9]{\frac{\sqrt[5]{x^2}\cdot\sqrt[5]{x^2}\cdots(100 \text{ veces})}{\sqrt[4]{x}\cdot\sqrt[4]{x}\cdots(340 \text{ veces})}} \right\}$$

$$-\frac{x^{-8}}{8}+C$$

Integral #4 Integratión Bee Physic

$$\int \frac{1}{1+\sqrt{x+1}}\,\mathrm{d}x$$

Respuesta Integral #4

Integratión Bee Physic

$$\int \frac{1}{1+\sqrt{x+1}}\,\mathrm{d}x =$$

$$2\sqrt{1+x} - 2\ln\left(1+\sqrt{1+x}\right) + C$$

Integral #33 **Integratión Bee Physic**

$$\int \frac{1}{x \ln(x) \ln(\ln x)} \, \mathrm{d}x$$

$$\int \frac{1}{x \ln(x) \ln(\ln x)} \mathrm{d}x =$$

$$\ln[\ln(\ln x)] + C$$

Integral #25 **Integratión Bee Physic**

$$\int \frac{x}{\sqrt{1-x^2}} \ln\left[\frac{x+1}{x-1}\right] \mathrm{d}x$$

Respuesta Integral #25

Integratión Bee Physic

$$\int \frac{x}{\sqrt{1-x^2}} \ln\left[\frac{x+1}{x-1}\right] \mathrm{d}x =$$

$$-\sqrt{1-x^2} \ln\left(\frac{x+1}{x-1}\right) + 2\arcsin(x) + C$$

Integral #55 **Integratión Bee Physic**

$$\int_0^2 [x^2]\,\mathrm{d}x$$

Respuesta Integral #55

Integratión Bee Physic

$$\int_0^2 [x^2]\,\mathrm{d}x = 5 - \sqrt{3} - \sqrt{2}$$